

EMILIANO DE LA GARZA VILLARREAL

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EDUCATION

University of Tennessee - Chattanooga
Bachelor of Science (BS), Data Science

August 2024 - May 2026
GPA: 3.8

Cleveland State Community College
Associate of Science (AS), Computer Science

August 2022 - December 2023
GPA: 3.7

PROFESSIONAL EXPERIENCE

Center of Urban Informatics and Progress
Machine Learning Intern

Chattanooga, TN, USA
April 2025 - Present

- Built a real-time intersection monitoring system on NVIDIA Jetson (Embedded System) using ROS 2; optimized performance by architecting an asynchronous multi-worker pipeline to maximize GPU utilization and eliminate thread-blocking latency for trajectory visualization.
- Improved Average Displacement Error (ADE) and Final Displacement Error (FDE) metrics by ~8% over baseline strategies by applying and combining FedProx and FedAdam within a Flow-based trajectory prediction model.
- Managed the applied machine learning lifecycle for a crash detection pipeline using Multiscale Vision Transformers (MViTv2), encompassing video dataset curation, spatio-temporal transformations, and local model training.

Eaton Corporation
Engineering Intern

Cleveland, TN, USA
December 2023 - Present

- Analyzed large-scale industrial datasets to identify operational trends and inefficiencies, translating raw metrics into actionable insights to generate over \$300,000 in annual savings.
- Conducted shop-floor time studies to build a predictive production model, updating system standards that reduced labor hours by 20%.

PROJECTS & OUTSIDE EXPERIENCE

Kaggle: March Machine Learning Mania Engine

- Developed a chronological Elo rating system for 40+ years of NCAA data, utilizing a Margin of Victory (MOV) multiplier and seasonal mean-reversion to capture team momentum.
- Built a PyTorch Multi-Layer Perceptron (MLP) leveraging CUDA-enabled GPU acceleration for high-throughput inference across thousands of tournament permutations, employing Isotonic Regression to calibrate probabilities.
- Designed a symmetrical data pipeline to eliminate team-order bias by training on relational differentials, such as the rate of change of Elo, rather than static seasonal averages.

Swarm-Sense (ROS 2 & Distributed Systems)

- Architected a high-performance ROS 2 Humble fleet interface to monitor 5+ robots simultaneously via Foxglove Bridge and Server-Client WebSockets.
- Reduced video latency to sub-100ms by implementing hardware-accelerated H.264 transport and 500kbps bitrate caps for bandwidth-constrained environments.
- Developed a custom TypeScript/React extension for edge-side telemetry caching and automated data export to MATLAB (.mat) formats.

SKILLS

Skills: Pytorch, Python, TensorRT, OpenCV, Linux/Unix, Git, ROS2, YOLO, RTSP, Transfer Learning, Data Augmentation, Federated Learning, Flowchain Model, FedProx, FedAdam, Time-Series Analysis, Asynchronous Multi-Worker Pipeline, Real-Time Data Pipelines, Visualization Techniques, Operational Analysis, Shop-Floor Time Studies, NVIDIA Jetson, Java, Pandas, NumPy, Machine Learning, Neural Networks, Data Engineering, Data Analysis, Computer Vision, Data Structures & Algorithms, C/C++

Languages: Spanish, English